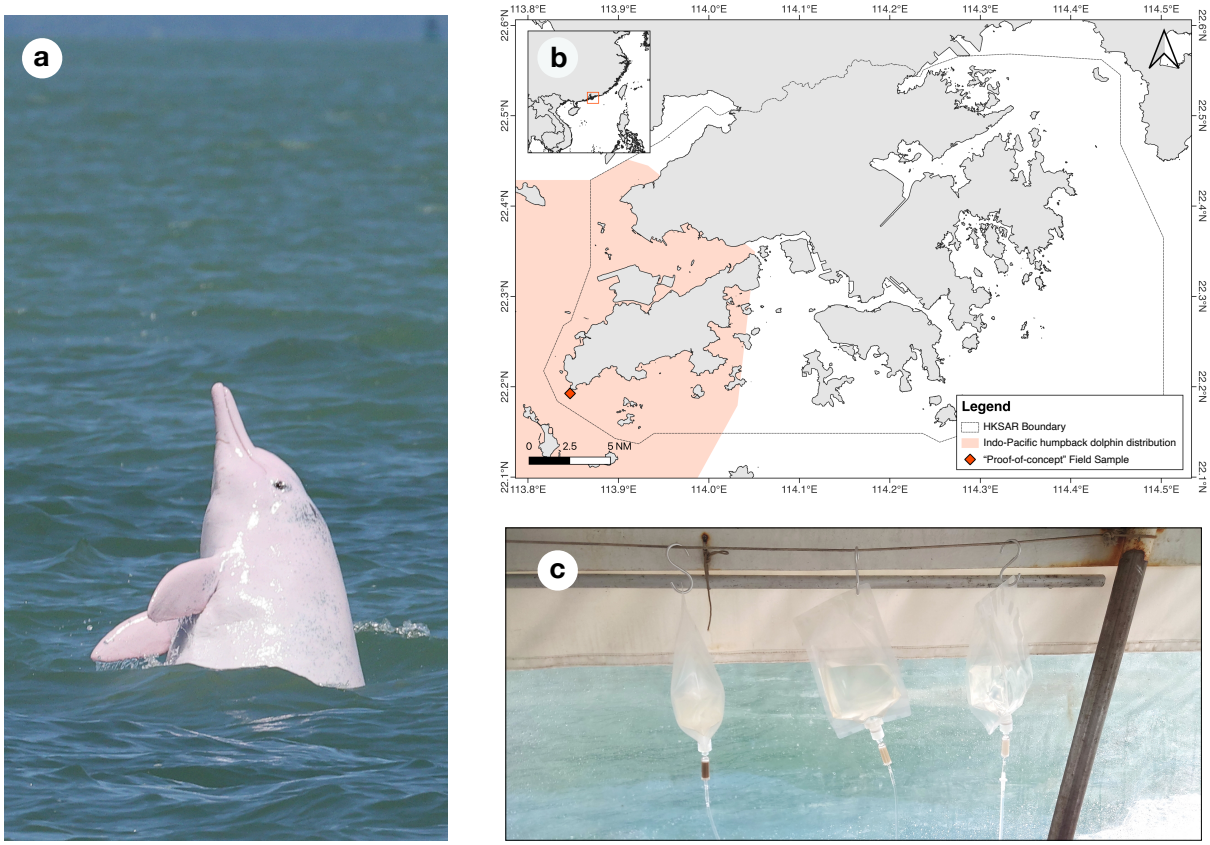


Figure 1 | Indo-Pacific humpback dolphin in Hong Kong waters (**a**) Indo-Pacific humpback dolphin (*Sousa chinensis*), or Chinese white dolphin, in Hong Kong waters (Photo taken by E. Matrai). (**b**) The distribution of *S. chinensis* around the Lantau island in Hong Kong waters. The red diamond marks the location where we sighted *S. chinensis* and collected the "proof-of-concept" field samples for qPCR analysis. (**c**) Gravity filtration of seawater on a boat during the field survey (Photo taken by M. Ushio). The photos are published under the CC BY 4.0 license.

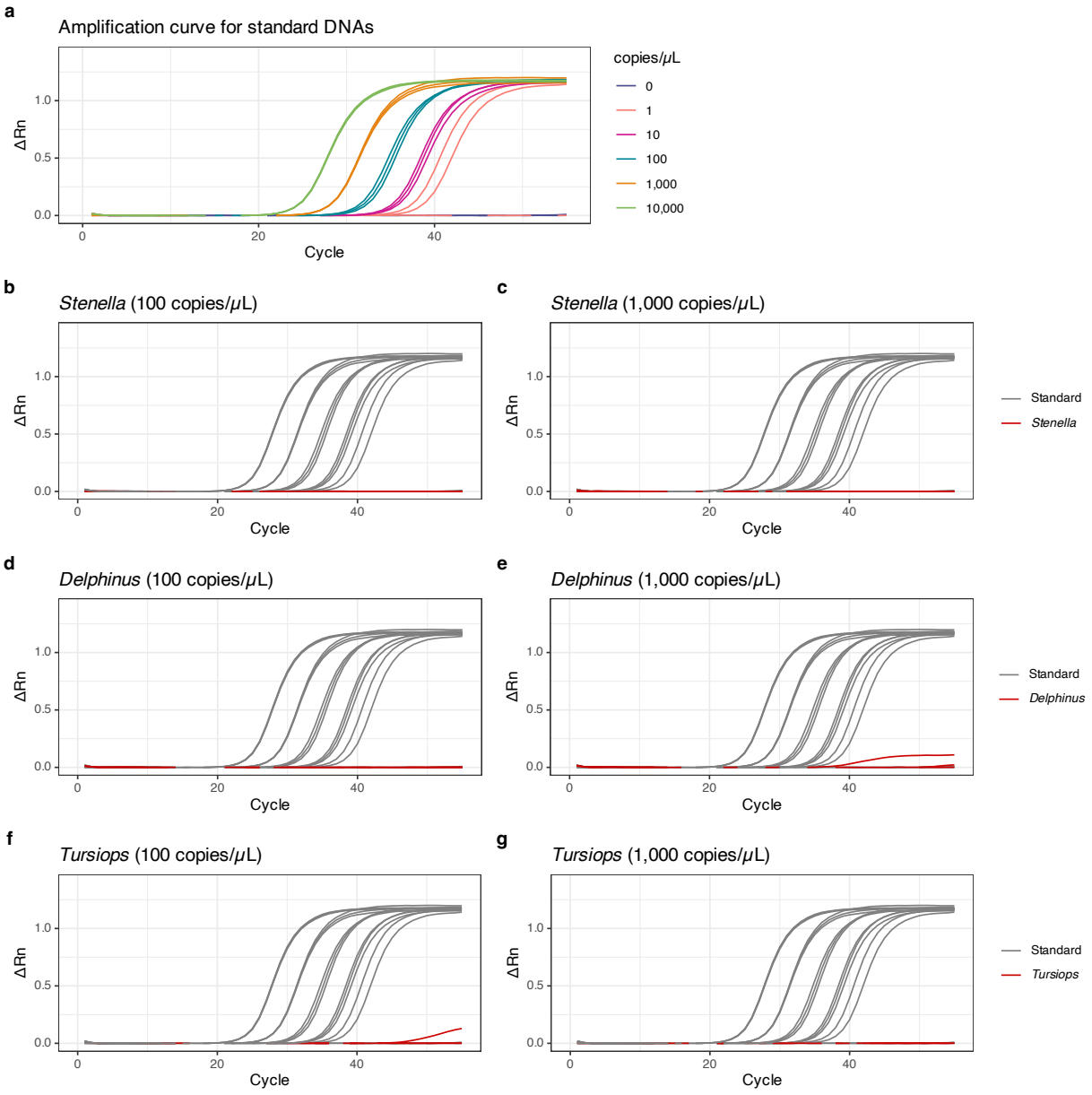


Figure 2 | Specificity tests of the *Sousa chinensis* qPCR assay against three sympatric Delphinidae species. (**a**) Amplification curves of *S. chinensis* standard DNA across a dilution series (0 to 10,000 copies/μL). Different colors indicate different template concentrations. (**b–g**) Amplification curves of synthesized target-region (Cytb) DNA from three non-target, but potentially co-occurring cetacean species (red curves) tested at 100 copies/μL (**b**, **d**, **f**; slightly exceeding the maximum concentration observed in our "proof-of-concept" field samples, ~60 copies/μL) and 1,000 copies/μL (**c**, **e**, **g**), plotted alongside the *S. chinensis* standard DNA curves (gray curves, 0 to 10,000 copies/μL): (**b**, **c**) *Stenella coeruleoalba*, (**d**, **e**) *Delphinus capensis* and (**f**, **g**) *Tursiops truncatus*.

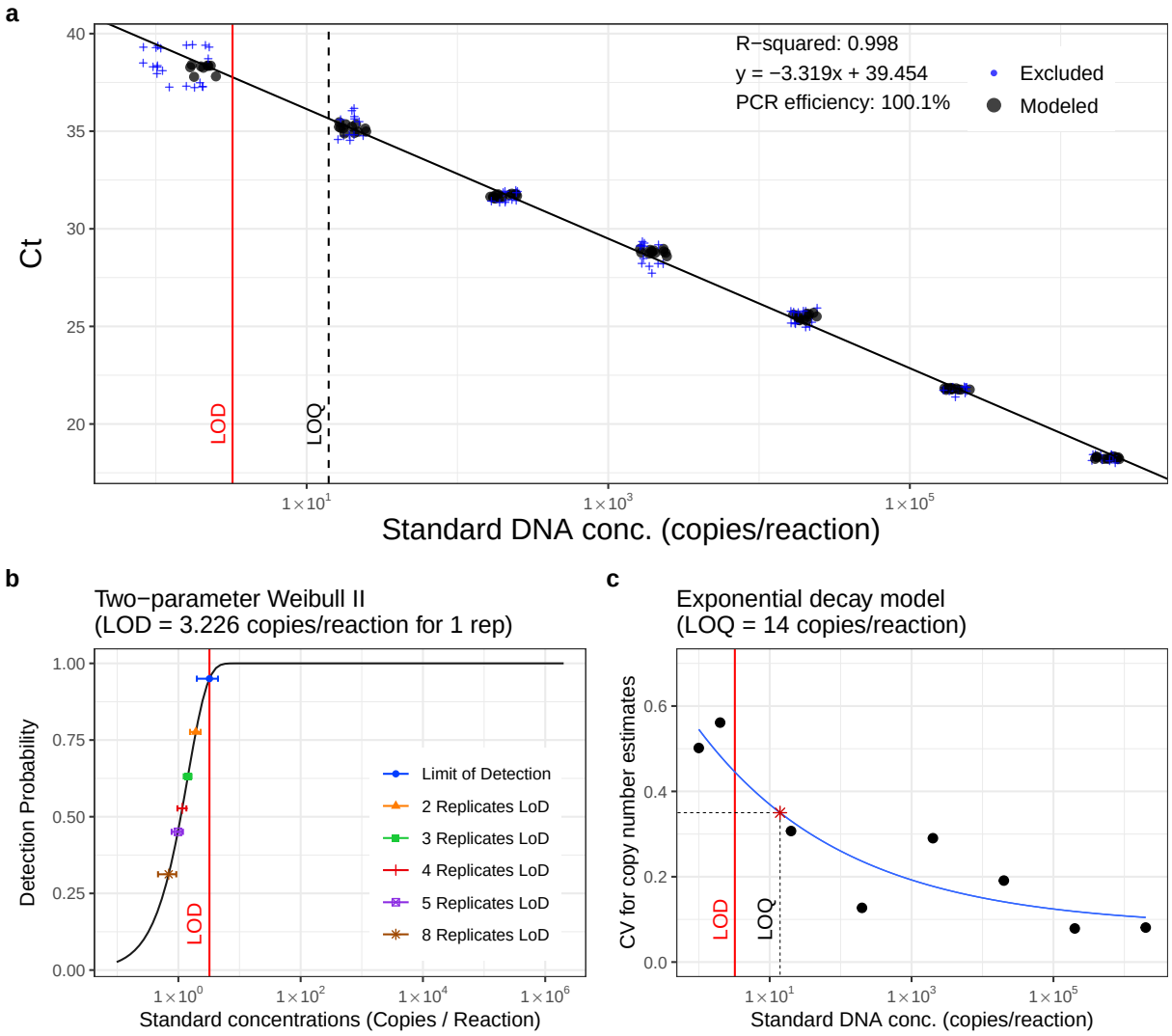


Figure 3 | Standard curve, the limit of detection (LOD) and limit of quantification (LOQ) of the primers and probe specific to *Sousa chinensis*. (**a**) Standard curve generated from 24 technical replicates across eight standard DNA concentrations (1 to 2.0 $\times$ 10^6 copies/reaction). The x-axis represents copy numbers of the *Cytb* target region per qPCR reaction, and the y-axis represents the C_t value. Following Klymus et al. (2020), filled black points indicate data included in the linear regression (the middle two quartiles, i.e., 25th–75th percentiles, of standards with $\geq$ 50% detection rates), whereas blue "+" symbols indicate data excluded from the regression (points outside the middle two quartiles or standards with $<$ 50% detection rates, i.e., 1 copy/reaction). The red solid vertical line and black dashed vertical line indicate the modeled LOD (3.22 copies/reaction) and modeled LOQ (14 copies/reaction), respectively. (**b**) Determination of the LOD. Open circles indicate observed detection rates at each standard concentration. The two-parameter Weibull type II model (`W2.2()` in R) was selected as the best-fitting model. Colored symbols with horizontal error bars indicate effective LODs (with 95% confidence intervals) for different numbers of technical replicates (1 to 8 replicates). (**c**) Determination of the LOQ based on the exponential decay model. The x-axis represents copy numbers and the y-axis represents the coefficient of variation (CV) for copy number estimates calculated following Forootan et al. (2017) and Klymus et al. (2020). The blue curve indicates the exponential decay model (see Fig. S2 and Table S7 for comparisons of all candidate models). The red vertical line indicates the modeled LOD, and the black "*" symbol indicates the 35% CV precision threshold, at which the copy number is defined as the modeled LOQ (14 copies/reaction, equivalent to 7.0 copies/μL).

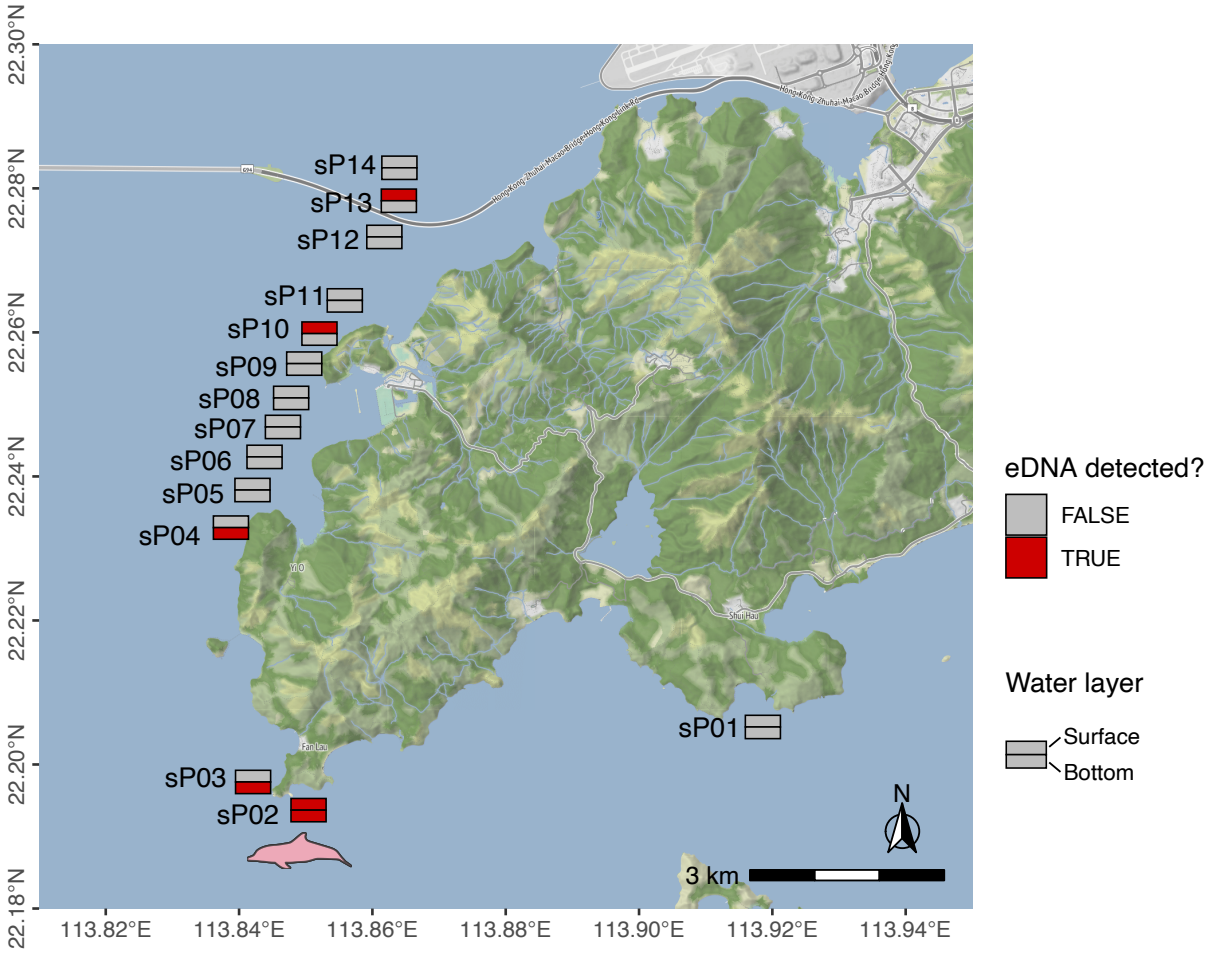


Figure 4 | Detection of *Sousa chinensis* eDNA from field seawater samples using qPCR. qPCR-based detection of *S. chinensis* eDNA from surface and bottom water samples taken in July 2024. Labels indicate the sample collection locations (see the metadata in the supporting information). Red and gray color indicate positive and negative detection, respectively. Upper and lower rectangles indicate the result for surface and bottom water layers, respectively. *S. chinensis* silhouette indicates *S. chinensis* sighting during the field survey (silhouette credit, Chris huh, CC BY-SA, https://creativecommons.org/licenses/by-sa/3.0/). Map data by OpenMapTiles and OpenStreetMap contributors, under ODbL.

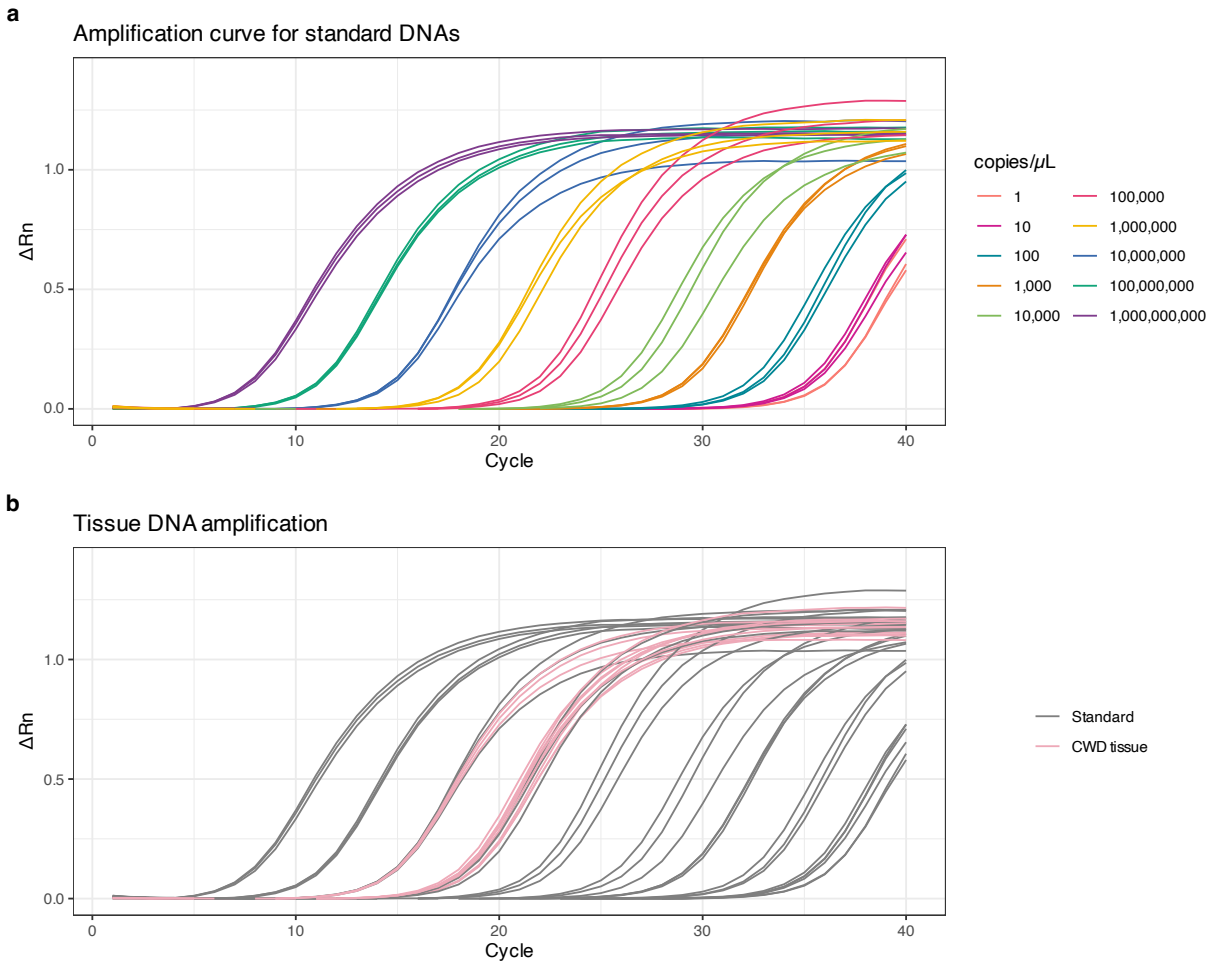


Figure S1 | Amplification of *Sousa chinensis* standard DNA and tissue-extracted DNA. (**a**) Amplification curve of the target region (*Cytb*) of *Sousa chinensis* (synthesized DNA). Different colors indicate different concentrations. Amplification curve of (**a**) S. chinensis standard DNAs and (**b**) *S. chinensis* tissue DNA (pink curve). Gray curves in **b** indicate amplification curves of different concentrations of *S. chinensis* standard DNA.

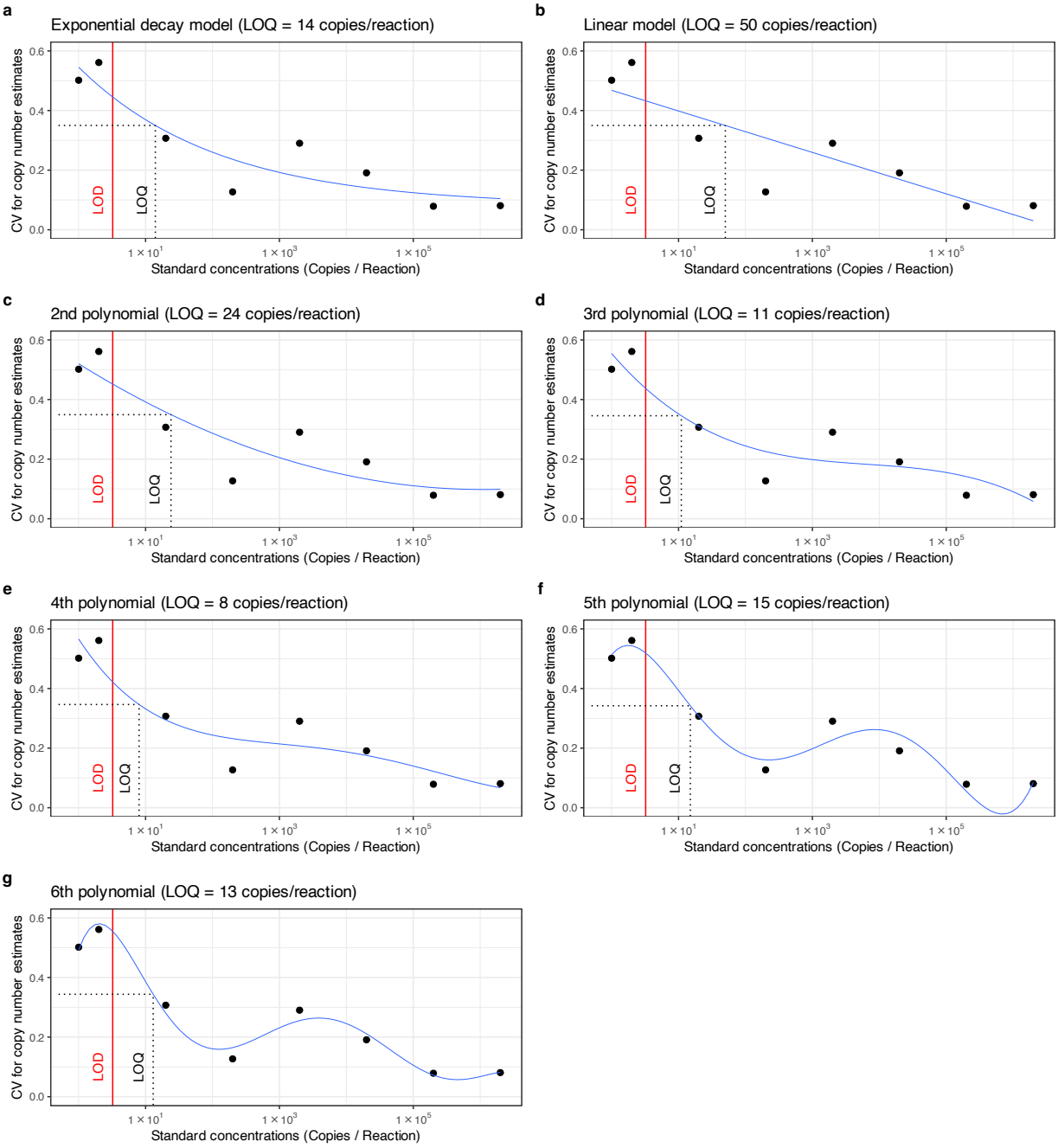


Figure S2 | Comparison of candidate models for determining the limit of quantification (LOQ) of the *Sousa chinensis* qPCR assay. (**a**) Exponential decay, (**b**) linear, and (**c–g**) 2nd- to 6th-order polynomial models. The x-axis represents standard DNA concentrations (copies/reaction), and the y-axis represents the coefficient of variation (CV) for copy number estimates calculated following Forootan et al. (2017) and Klymus et al. (2020). Black points indicate observed CVs at each standard concentration ($n = 24$ replicates), and blue curves indicate the fitted models. Red vertical lines indicate the modeled LOD (3.22 copies/reaction), and black dotted lines indicate the 35% CV threshold (horizontal) and the resulting modeled LOQ (vertical). Model diagnostics (RSE, AIC, AICc, and BIC) for all seven models are provided in Table S7.

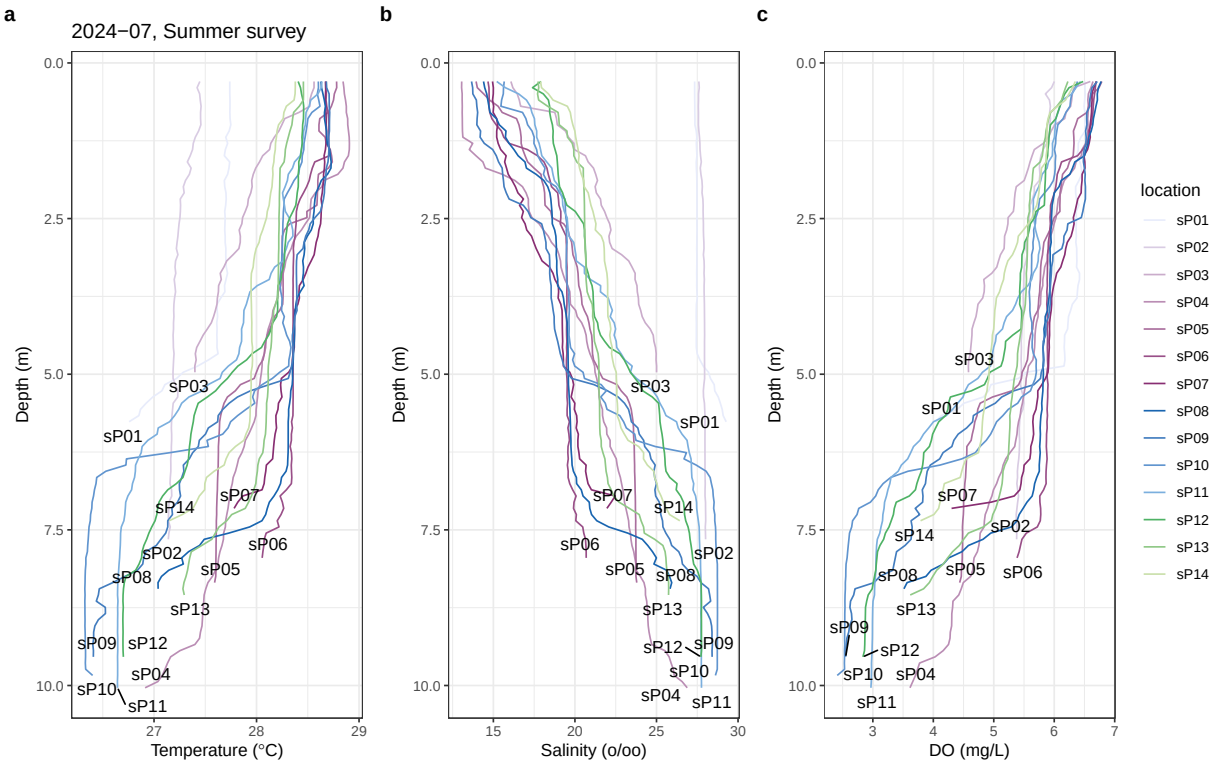


Figure S3 | Physical profiles of seawater at the sampling locations. (**a**) Temperature, (**b**) salinity, and (**c**) dissolved oxygen (DO) at each sampling location for the summer survey in July 2024. Different colors indicate different sampling locations, and labels indicate sampling locations.

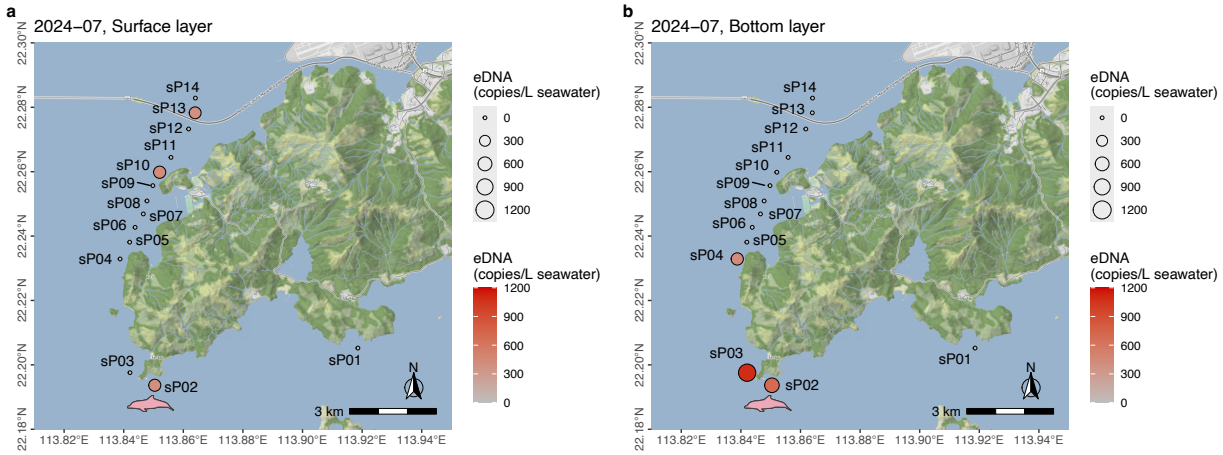


Figure S4 | Concentrations of *Sousa chinensis* eDNA from field seawater samples using qPCR. Concentrations of *S. chinensis* eDNA from (**a**) surface and (**b**) bottom water samples taken in July 2024. Labels indicate sample collection locations. Color density and size of each point represent eDNA concentration. Most the eDNA concentrations are below the limit of quantification (LOQ: 14 copies/reaction or 7 copies/μL) and are therefore unreliable, but the results are provided as supplementary information. *S. chinensis* silhouette indicates *S. chinensis* sighting during the field survey (silhouette credit, Chris huh, CC BY-SA, https://creativecommons.org/licenses/by-sa/3.0/). Map tiles by Stamen Design, hosted by Stadia Maps, under CC BY 4.0. Map data by OpenMapTiles and OpenStreetMap contributors, under ODbL.

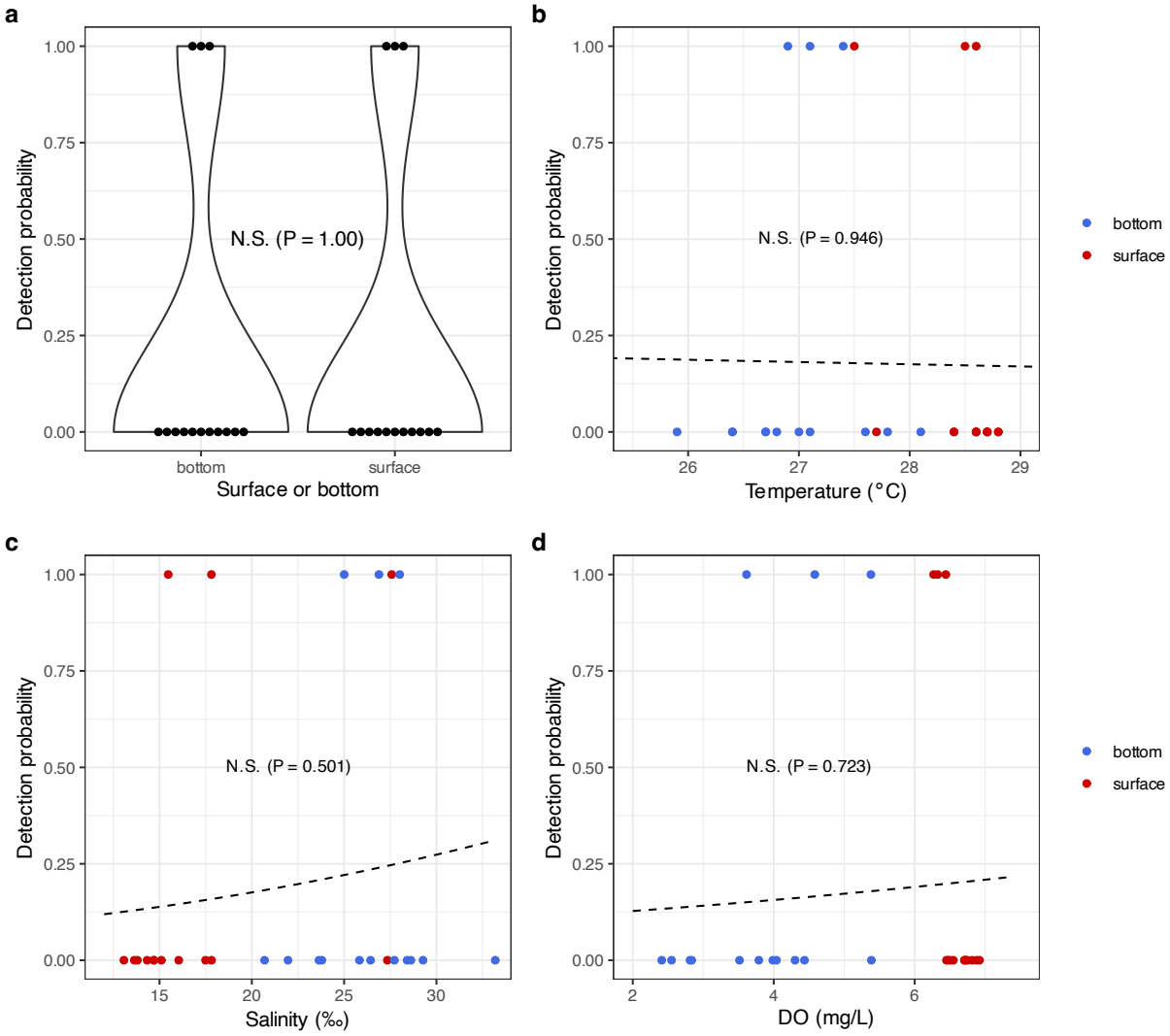


Figure S5 | The relationship between seawater physical properties and qPCR detection in July 2024. The relationship between qPCR detection of *Sousa chinensis* eDNA and (**a**) the origin of water samples (surface or bottom), (**b**) temperature, (**c**) salinity, and (**d**) dissolved oxygen (DO), respectively. Red and blue colors indicate surface and bottom water samples, respectively. Lines in **b-d** indicate model predictions based on binomial GLMM with each sampling location as a random factor (`glmer(detection ~ environmental variable + (1 | location), family = binomial)`), although all the regressions are not statistically significant. Note that only qualitative results were used for this analysis because the eDNA concentrations of most samples were below the limit of quantification (LOQ).